Probability from Observation:
Collective Exhaustion and the Observable Extension Theorem

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Abstract

We prove that a coherent family of observations determines a unique probability measure on the observable σ-algebra. The setting is a directed system of measurable outcome spaces with surjective refinement maps and a compatible family of finitely additive charges on the cylinder algebra. The result is illustrated by two constructions.

A *direct Carathéodory construction* assumes compatible σ-additive marginals and builds the global measure directly without topological assumptions. We then relax σ-additivity to finite additivity and regain later via extension under a support condition.

The necessary and sufficient condition for σ-additive extension is *collective exhaustion* (CE): no level can assign persistent mass to events the system as a whole sees as empty. The fact that CE is not derivable from any structural condition on the query system is precisely located by the finite-cofinite counterexample and a model-theoretic argument via Łoś's theorem.

A *Stone construction* accordingly assumes only finite additivity leveraging exhaustion to extend: compactness of the Stone space manufactures σ-additivity on the compact completion where CE is the support condition that descends the measure back to the space. In limit-descent exhaustion acquires geometric form via natural completion.

We take the Stone space $\mathrm{St}(\mathcal{C})$ as the canonical compact Hausdorff completion of the observable structure: the space of all consistent distinction patterns extending the cylinder algebra. The two constructions differ by where the missing σ-additive structure is supplied — internally by countable reach in the directed system, or externally by the Stone completion — and whenever both apply, CE ensures they produce the same observable measure.

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1 Introduction

Observable distinctions are classically typified by Boolean algebras. Their study crucially underpins modern measure theory and its probabilistic extensions. In contrast, the directed nature of observation, wherein finer levels of resolution refine coarser ones, has been relatively misprized toward such ends. In the foundational treatments of Kolmogorov [7] and Choksi [3], the global σ-algebra emerges as the projective limit of compatible measure spaces. Directed structure plays an organisational role: it orders the marginals and licenses the assembly, but is not itself interrogated as a source of probabilistic constraint.

We facilitate interrogation by a directed system of Boolean algebras connected by surjective refinement maps. Valuations enter as compatible families of finitely additive charges on the level algebras: charges at different levels agree on events seen from both. Structural coherence and valuation compatibility together constitute the finitary data.

Under such treatment Kolmogorov's question resurfaces without dependence on the ontological status of a global state space. In the language of *queries*: what further condition on valuations is required for finitary data to support a genuine probability measure on the observable σ-algebra?

The condition is *collective exhaustion* (CE): a compatible family of finitely additive charges extends to a σ-additive measure if and only if, whenever a decreasing sequence of events vanishes globally, there exists a level at which the sequence's mass converges to zero. CE rules out purely finitely additive charges, which assign persistent mass to sequences that vanish in the limit. It is the exact admissibility condition separating observational coherence from genuine probability: a coherent family of charges deserves to be treated as a probability system precisely when it satisfies CE. A family satisfying CE determines a unique probability measure on the observable σ-algebra; no first-order structural condition can substitute for it.

Organisation. Section 2 introduces query systems, cylinder sets, the observable σ-algebra, compatible charges, and Surjective Evaluation. Section 3 gives the direct extension theorem: given compatible σ-additive marginals, the global observable measure follows by Carathéodory. Section 4 establishes CE as the admissibility criterion that determines when finitely additive level data warrants that probabilistic status, proves its non-derivability from structural conditions, and shows that the underlying logical obstruction is not specific to the query-system language (§4.5). Section 5 proves the Stone realization from finitely additive data, introducing Discriminability as the condition that makes the Stone embedding injective. Section 6 draws out CE's unified meaning across both constructions and characterises the Stone space. The main results have been formalized in Lean 4 using Mathlib; source files are available in the accompanying repository. This paper is the first step in a programme deriving canonical dynamics, state-space reconstruction, and certifiability from observational data; the further steps are carried out in the companion papers.

2 Query systems and charges

2.1 Observational structure

Rather than a single σ-algebra of events, we work with a directed system of Boolean algebras representing observable distinctions at increasing resolution; the global σ-algebra arises as their completion.

Definition 2.1 (Observational structure)**.** An *observational structure* consists of:

- a nonempty partially ordered set $(\iota, \leq)$, the *index set*;
- for each $i \in \iota$, a measurable space $(\mathsf{O}_i, \mathcal{O}_i)$, the *outcome space at level* i;
- for each $i \leq j$ in ι, a measurable surjection $\pi_{ij} : \mathsf{O}_j \to \mathsf{O}_i$, the *refinement map*,

satisfying $\pi_{ik} = \pi_{ij} \circ \pi_{jk}$ for all $i \leq j \leq k$.

Definition 2.2 (Instantiation; query system)**.** An *instantiation* of an observational structure $(\iota, \leq, \{(\mathsf{O}_i, \mathcal{O}_i)\}, \{\pi_{ij}\})$ is a set Ω together with maps $\mathrm{eval}_i : \Omega \to \mathsf{O}_i$ for each $i \in \iota$, the *evaluation maps*, satisfying the *coherence condition* $\pi_{ij} \circ \mathrm{eval}_j = \mathrm{eval}_i$ for all $i \leq j$. A *query system* is an observational structure together with an instantiation.

The index set, outcome spaces, and refinement maps encode the distinctions the observer can make and how finer levels coarsen. Choosing Ω is an additional act: it is a set of states realising the abstract structure concretely, and coherence says evaluation is compatible with coarsening.

2.2 Canonical instantiation and observables

Every observational structure admits a canonical instantiation: the space of all coherent assignments of outcomes across levels. This identifies Ω with the set of all globally consistent observation patterns. Explicitly, the canonical instantiation is the projective limit

$$\Omega = \varprojlim_{i\in\iota} \mathsf{O}_i = \Big\{\omega \in \prod_{i\in\iota} \mathsf{O}_i \;\Big|\; \pi_{ij}(\omega_j) = \omega_i \text{ whenever } i \leq j\Big\},$$

with $\mathrm{eval}_i(\omega) = \omega_i$ and $\omega \in \Omega$ generic.

Example 2.3 (Bernoulli observations)**.** Let ι be the set of finite subsets of $\mathbb{N}$, ordered by inclusion. For $i \in \iota$, set $\mathsf{O}_i = \{0,1\}^{|i|}$ with its discrete σ-algebra $\mathcal{O}_i$, and for $i \subseteq j$ let $\pi_{ij} : \mathsf{O}_j \to \mathsf{O}_i$ be the coordinate projection. An instantiation assigns to each ω a consistent family of finite binary observations; the canonical instantiation identifies Ω with $\{0,1\}^{\mathbb{N}}$. Cylinder sets correspond to fixing finitely many coordinates, and a compatible family $\{\nu_i\}$ is precisely the system of finite-dimensional distributions of a Bernoulli process.

Definition 2.4 (Common coarsenings)**.** The index set $(\iota, \leq)$ admits *common coarsenings* if every pair $i, j \in \iota$ has a lower bound: there exists $k \leq i$ and $k \leq j$.

Definition 2.5 (Cylinder sets)**.** For $i \in \iota$ and $A \in \mathcal{O}_i$, the *level-i cylinder with base A* is

$$\mathrm{Cyl}(i, A) \;=\; \{\omega \in \Omega : \mathrm{eval}_i(\omega) \in A\}.$$

The *cylinder family* is $\mathcal{C} = \{\mathrm{Cyl}(i, A) : i \in \iota,\ A \in \mathcal{O}_i\}$, and the *observable σ-algebra* is $\sigma(\mathcal{C})$.

Lemma 2.6 (Cylinder π-system)**.** *Assume $(\iota, \leq)$ admits common coarsenings. Then $\mathcal{C}$ is a π-system: it is closed under finite intersections.*

Proof. The coherence condition gives $\mathrm{Cyl}(i, A) = \mathrm{Cyl}(j, \pi_{ij}^{-1}(A))$ for $i \leq j$: the same subset of Ω is a cylinder at any finer level. Given $\mathrm{Cyl}(i, A)$ and $\mathrm{Cyl}(j, B)$, let k be a common lower bound. Then

$$\mathrm{Cyl}(i, A) \cap \mathrm{Cyl}(j, B) = \mathrm{Cyl}(k,\ \pi_{ki}^{-1}(A) \cap \pi_{kj}^{-1}(B)),$$

which is again a cylinder. □

For each $i \in \iota$, let B_i denote the Boolean algebra of level-i cylinders $\{\mathrm{Cyl}(i, A) : A \in \mathcal{O}_i\}$; then $\mathcal{C} = \bigcup_{i\in\iota} B_i$. For $i \leq j$, define the *connecting map*

$$\varphi_{ij} : B_i \to B_j, \qquad \varphi_{ij}(\mathrm{Cyl}(i, A)) = \mathrm{Cyl}(j, \pi_{ij}^{-1}(A)).$$

This is a well-defined Boolean algebra homomorphism. The family $\{B_i, \varphi_{ij}\}$ is a directed system of Boolean algebras.

2.3 Compatible charges

Definition 2.7 (Compatible family)**.** A *charge* on a Boolean algebra B is a finitely additive function $\mu : B \to [0, \infty)$ with $\mu(\emptyset) = 0$. A family of charges $\{\mu_i\}_{i\in\iota}$ with μ_i on B_i is *compatible* if

$$\mu_j(\varphi_{ij}(E)) = \mu_i(E) \qquad \text{for all } i \leq j \text{ and all } E \in B_i.$$

A compatible family also determines a charge on $\mathcal{C}$ by $\mu(\mathrm{Cyl}(i, A)) := \mu_i(\mathrm{Cyl}(i, A))$; compatibility ensures this is well-defined.

Both constructions require the following condition on the query system.

Definition 2.8 (Surjective Evaluation)**.** The query system satisfies *Surjective Evaluation* if $\mathrm{eval}_i : \Omega \to \mathsf{O}_i$ is surjective for every $i \in \iota$.

Surjective evaluation ensures every outcome at every level is realised by some $\omega \in \Omega$; it implies the refinement maps π_{ij} are surjective and the connecting maps φ_{ij} injective.

3 Direct extension from compatible probability marginals

Given compatible σ-additive probability marginals, Carathéodory's theorem yields a global measure directly on the instantiation space. The input is a query system satisfying surjective evaluation, sequential common refinements, and common coarsenings, together with a compatible family of σ-additive probability marginals.

3.1 Observational determination

Proposition 3.1 (Observational determination)**.** *Let the query system admit common coarsenings and let P, P' be probability measures on $(\Omega, \sigma(\mathcal{C}))$. If $(\mathrm{eval}_i)_\# P = (\mathrm{eval}_i)_\# P'$ for all $i \in \iota$, then $P = P'$.*

Proof. The equality of pushforward laws implies P and P' agree on every cylinder event. By Lemma 2.6 the cylinder family $\mathcal{C}$ is a π-system generating $\sigma(\mathcal{C})$. The π–λ theorem gives $P = P'$. □

3.2 Cylinder content

Definition 3.2 (Common refinements)**.** The index set $(\iota, \leq)$ admits *common refinements* if every pair $i, j \in \iota$ has an upper bound: there exists $k \geq i$ and $k \geq j$.

The cylinder family may represent the same event at multiple levels: $\mathrm{Cyl}(i, A)$ and $\mathrm{Cyl}(j, B)$ coincide in Ω whenever A and B are preimages of a common set at a finer level. Assigning a numerical value to such an event is well-defined only if the two representations agree — and verifying this requires passing to a common refinement of i and j.

Lemma 3.3 (Finite observational content)**.** *Assume the query system admits common coarsenings and common refinements and satisfies Surjective Evaluation, and let $\{\nu_i\}_{i\in\iota}$ be a family with each $\nu_i \in \mathcal{P}(\mathsf{O}_i)$ compatible under the refinement maps, i.e. $(\pi_{ij})_\# \nu_j = \nu_i$ for all $i \leq j$. Then the assignment*

$$\mu_0(\mathrm{Cyl}(i, A)) := \nu_i(A), \qquad A \in \mathcal{O}_i,$$

defines a well-defined finitely additive content on $\mathcal{C}$.

Proof. Well-definedness. Suppose $\mathrm{Cyl}(i, A) = \mathrm{Cyl}(j, B)$. Use common refinements to find k with $k \geq i$ and $k \geq j$. The constraint sets in O_k are $\pi_{ik}^{-1}(A)$ and $\pi_{jk}^{-1}(B)$. The cylinder equality and surjective evaluation (surjectivity of eval_k) imply these coincide in O_k. Compatibility then gives $\nu_i(A) = \nu_k(\pi_{ik}^{-1}(A)) = \nu_k(\pi_{jk}^{-1}(B)) = \nu_j(B)$.

Finite additivity. For disjoint cylinders at possibly different levels, compress to a common refinement k and use additivity of ν_k. □

3.3 Carathéodory extension theorem

Definition 3.4 (Sequential common refinements)**.** The index set $(\iota, \leq)$ admits *sequential common refinements* if every countable family $\{i_n\}_{n\geq 1} \subset \iota$ has an upper bound in ι. Sequential common refinements imply common refinements.

Finite additivity of the cylinder content is not sufficient for Carathéodory's theorem: one needs σ-subadditivity, which requires compressing a countable cylinder cover $\{\mathrm{Cyl}(i_n, A_n)\}$ to a single level. Common refinements handle finite families; sequential common refinements extend this to countable ones, supplying the index-set reach the proof requires.

Theorem 3.5 (Carathéodory observational extension)**.** *Suppose:*

(i) $(\iota, \leq)$ admits common coarsenings and sequential common refinements;

(ii) the query system satisfies Surjective Evaluation;

(iii) each ν_i is a probability measure on $(\mathsf{O}_i, \mathcal{O}_i)$, and the family is compatible under the refinement maps: $(\pi_{ij})_\# \nu_j = \nu_i$ for all $i \le j$.

Then there exists a unique probability measure P on $(\Omega, \sigma(\mathcal{C}))$ such that $(\mathrm{eval}_i)_\# P = \nu_i$ for every $i \in \iota$.

Proof. Step 1: Finite content. Theorem 3.3 gives a well-defined finitely additive content μ_0 on $\mathcal{C}$.

Step 2: σ-subadditivity. Let $\mathrm{Cyl}(i, B) \subseteq \bigcup_{n \ge 1} \mathrm{Cyl}(i_n, A_n)$. Apply sequential common refinements to $\{i, i_1, i_2, \ldots\}$ to find a single index k with $k \ge i$ and $k \ge i_n$ for all n. Compress each cylinder to level k:

$$\mathrm{Cyl}(i, B) = \mathrm{Cyl}(k, E), \qquad \mathrm{Cyl}(i_n, A_n) = \mathrm{Cyl}(k, E_n),$$

where $E = \pi_{ik}^{-1}(B)$ and $E_n = \pi_{i_n k}^{-1}(A_n)$. Surjective evaluation and the cylinder inclusion in Ω imply $E \subseteq \bigcup_n E_n$ in O_k. Therefore

$$\mu_0(\mathrm{Cyl}(i, B)) = \nu_k(E) \le \sum_{n=1}^{\infty} \nu_k(E_n) = \sum_{n=1}^{\infty} \mu_0(\mathrm{Cyl}(i_n, A_n)),$$

using σ-subadditivity of the single measure ν_k.

Step 3: Carathéodory extension. The family $\mathcal{C}$ is a set semiring generating $\sigma(\mathcal{C})$. It is closed under finite intersections by Lemma 2.6. Moreover, if $E = \mathrm{Cyl}(i, A)$ and $F = \mathrm{Cyl}(j, B)$, choose k with $k \ge i, j$; then

$$E = \mathrm{Cyl}\big(k,\, \pi_{ik}^{-1}(A)\big), \qquad F = \mathrm{Cyl}\big(k,\, \pi_{jk}^{-1}(B)\big),$$

so

$$E \setminus F = \mathrm{Cyl}\big(k,\, \pi_{ik}^{-1}(A) \setminus \pi_{jk}^{-1}(B)\big) \in \mathcal{C}.$$

Thus $\mathcal{C}$ is closed under relative differences, confirming the semiring property. Since μ_0 is a finitely additive content on $\mathcal{C}$ that is σ-subadditive for countable covers by cylinder sets (Step 2), the semiring form of Carathéodory's extension theorem yields a measure P on $(\Omega, \sigma(\mathcal{C}))$ extending μ_0 (Bogachev [1], Vol. I, Corollary 1.11.9).

Step 4: Marginal recovery and normalization. For any $A \in \mathcal{O}_i$, $P(\mathrm{Cyl}(i, A)) = \mu_0(\mathrm{Cyl}(i, A)) = \nu_i(A)$, so $(\mathrm{eval}_i)_\# P = \nu_i$. Setting $A = \mathsf{O}_i$ gives $P(\Omega) = 1$.

Uniqueness. Two probability measures with the same evaluation marginals agree on $\mathcal{C}$, hence on $\sigma(\mathcal{C})$ by Theorem 3.1. ☐

Remark 3.6 (Relation to Kolmogorov's extension theorem)**.** Kolmogorov's theorem [7] constructs a probability measure on S^T from consistent finite-dimensional distributions. Theorem 3.5 plays the same role for query systems: the instantiation $\Omega = \varprojlim \mathsf{O}_i$ replaces the product S^T, compatible marginals replace consistent finite-dimensional distributions, and sequential common refinements replace the countable-cofinality condition. The key conceptual difference is that no topology is imposed on the outcome spaces: the measure P is built directly on the measurable structure of Ω by Carathéodory, without any appeal to tightness or compactness.

4 Collective Exhaustion

Section 3 showed that compatible σ-additive marginals extend directly to a global observable measure. This section identifies the admissibility condition that determines when finitely additive level data deserves that probabilistic status.

Definition 4.1 (Collective Exhaustion)**.** A charge μ on $\mathcal{C}$ (the common extension of a compatible family $\{\mu_i\}$) satisfies *collective exhaustion* (CE) if: whenever $E_1 \supseteq E_2 \supseteq \cdots$ in $\mathcal{C}$ and $\bigcap_n E_n = \emptyset$, we have $\mu(E_n) \to 0$.

CE is a valuation-layer condition: it constrains how charges behave, not the Boolean-algebraic index structure. The following example shows it is not automatic.

Example 4.2 (Finite–cofinite obstruction)**.** Let $\mathbb{N}$ carry the finite–cofinite algebra $\mathcal{E}$ where $A \in \mathcal{E}$ if A is finite or A^c is finite. Define a normalized charge by $\mu_0(A) = 0$ if A is finite and $\mu_0(A) = 1$ if A is cofinite.

Disjoint sets in $\mathcal{E}$ are either both finite or one finite and one cofinite (two disjoint cofinite sets cannot exist), so finite additivity holds.

The sequence $E_n = \{n, n+1, \ldots\}$ satisfies $E_n \downarrow \varnothing$, yet each E_n is cofinite, so $\mu_0(E_n) = 1$ for all n: mass is not exhausted. No σ-additive extension exists.

We work first at the level of an abstract directed system of Boolean algebras, writing $\mathcal{E}_i$ for the algebra at level i and ℓ_i for its charge; the query-system interpretation returns in §5.

4.1 Single-algebra extension

Definition 4.3 (Boolean charge space)**.** A *Boolean charge space* is a pair $(\mathcal{E}, \ell)$ where $\mathcal{E}$ is a Boolean algebra of subsets of a set O and $\ell : \mathcal{E} \to [0, 1]$ is a normalized finitely-additive valuation. No σ-algebra and no topology are assumed on O.

By Stone's representation theorem [11], $\mathcal{E}$ is isomorphic to the clopen algebra of a compact totally-disconnected Hausdorff space $S(\mathcal{E})$, and $\sigma(\mathcal{E})$ is the Baire σ-algebra of $S(\mathcal{E})$. The sets in $\sigma(\mathcal{E})\backslash\mathcal{E}$ are Baire events that countable operations can reach but no single observable distinction can name.

Lemma 4.4 (Extension criterion)**.** *A normalized finitely-additive $\ell : \mathcal{E} \to [0, 1]$ extends to a σ-additive measure on $\sigma(\mathcal{E})$ if and only if ℓ is continuous at $\emptyset$: for every decreasing sequence $E_1 \supseteq E_2 \supseteq \cdots$ in $\mathcal{E}$ with $\bigcap_n E_n = \emptyset$ in $\sigma(\mathcal{E})$, we have $\ell(E_n) \to 0$. The extension is then unique.*

Proof. Standard; see Halmos [5] or Fremlin [4] Vol. 1. Continuity at $\emptyset$ is used in the Carathéodory extension argument to pass from finite additivity to σ-additivity on $\sigma(\mathcal{E})$. $\square$

The condition in Lemma 4.4 is not verifiable within $\mathcal{E}$ alone: confirming $\bigcap_n E_n = \emptyset$ requires access to $\sigma(\mathcal{E})$. In a directed system, one might hope a finer level $j \geq i$ could witness the emptiness level i cannot see — but the same obstruction reappears: finite additivity at j does not force continuity at $\emptyset$ for sequences in $\sigma(\mathcal{E}_j)$. Section 4.2 shows the directed-system structure alone does not provide it.

4.2 Nontrivial refinement

Definition 4.5 (Nontrivial refinement)**.** Let $\pi : O_j \to O_i$ be a surjective refinement map, inducing the Boolean algebra homomorphism $\pi^{-1} : \mathcal{E}_i \to \mathcal{E}_j$. The refinement is *nontrivial* if there exists $A \in \mathcal{E}_j$ with $A \notin \pi^{-1}(\mathcal{E}_i)$.

Proposition 4.6 (Nontrivial refinement introduces new events)**.** *If $j \geq i$ is a nontrivial refinement, then $\pi^{-1}(\mathcal{E}_i) \subsetneq \mathcal{E}_j$. Equivalently, a level i at which $\mathcal{E}_j = \pi^{-1}(\mathcal{E}_i)$ for every $j \geq i$ is one above which the refinement order is algebraically trivial: no finer level adds new discriminative power.*

Proof. Nontriviality asserts the existence of $A \in \mathcal{E}_j$ with $A \notin \pi^{-1}(\mathcal{E}_i)$, directly giving the strict inclusion. The contrapositive is the stated equivalence. $\square$

Lemma 4.7 (Strict enlargement under σ-closure). *Let $\mathcal{E}$ be a Boolean algebra of subsets of a countably infinite set O that is not a σ-algebra. Then $\sigma(\mathcal{E}) \setminus \mathcal{E} \neq \emptyset$.*

Proof. If $\mathcal{E}$ is not a σ-algebra, there exists a countable family $\{A_n\} \subset \mathcal{E}$ with $\bigcup_n A_n \notin \mathcal{E}$. By definition $\bigcup_n A_n \in \sigma(\mathcal{E})$, so it lies in $\sigma(\mathcal{E}) \setminus \mathcal{E}$. □

4.3 CE characterises σ-additivity

The directed-system version of CE and the main equivalence are as follows.

Definition 4.8 (Collectively exhaustive family). A compatible family $\{\ell_i\}$ of normalized finitely-additive charges on a directed system is *collectively exhaustive* if: for every $i \in \mathcal{I}$ and every decreasing sequence $E_1 \supseteq E_2 \supseteq \cdots$ in $\mathcal{E}_i$ with $\bigcap_n E_n = \emptyset$, there exists $j \geq i$ such that $\ell_j(\pi_{ij}^{-1}(E_n)) \to 0$.

Here $\pi_{ij} : O_j \to O_i$ is the refinement map (j finer than i), so $\pi_{ij}^{-1}(E_n) \subseteq O_j$ is the preimage of E_n at the finer level. No level may assign persistent mass to events the system as a whole sees as empty.

The following proposition constructs the minimal directed system in which the index-layer hypotheses hold while the finite-cofinite charge still fails σ-additive extension. The construction is a deliberate stress test, not a naturally arising observational system.

Proposition 4.9 (Index-layer conditions do not force σ-additivity). *There exists a compatible family of normalized finitely-additive charges on a sequentially upper-directed directed system that fails σ-additive extension at every level.*

Proof. We take the simplest sequentially upper-directed index set: adjoin a single terminal element ω to $\mathbb{N}$, setting $n \leq \omega$ for all $n \in \mathbb{N}$. Take all outcome spaces to be $\mathbb{N}$, all event algebras to be the finite-cofinite algebra $\mathcal{E}$, all refinement maps to be the identity, and each ℓ_i to be the finite-cofinite charge ℓ of Example 4.2. Compatibility holds trivially. The system satisfies every structural condition. By Example 4.2, ℓ is not σ-additive and admits no σ-additive extension; hence every level fails σ-additive extension. □

Theorem 4.10 (CE Characterisation). *Let $\{\ell_i\}$ be a normalized compatible family of finitely-additive charges on a directed system indexed by $\mathcal{I}$. The following are equivalent:*

(i) The family is collectively exhaustive.

(ii) ℓ_i extends uniquely to a σ-additive measure on $\sigma(\mathcal{E}_i)$ for every $i \in \mathcal{I}$.

Proof. (i) $\Rightarrow$ (ii). Fix $i \in \mathcal{I}$ and a decreasing sequence $E_n \searrow \emptyset$ in $\mathcal{E}_i$. By CE, find $j \geq i$ with $\ell_j(\pi_{ij}^{-1}(E_n)) \to 0$. By compatibility, $\ell_i(E_n) = \ell_j(\pi_{ij}^{-1}(E_n)) \to 0$. So ℓ_i is continuous at $\emptyset$. Lemma 4.4 gives the unique σ-additive extension.

(ii) $\Rightarrow$ (i). Suppose ℓ_i extends to μ_i for every i. Let $E_n \searrow \emptyset$ in $\mathcal{E}_i$. Then $\ell_i(E_n) = \mu_i(E_n) \to \mu_i(\emptyset) = 0$ by σ-additivity of μ_i and normalization. Take $j = i$. □

The question "do the structural conditions on a directed system force σ-additivity?" is ill-posed: it conflates conditions on the index structure with conditions on the charges. Proposition 4.9 separates them; Theorem 4.10 identifies the exact valuation-layer condition. Sequential upper-directedness enters only when assembling per-level extensions into a global measure (§3).

4.4 CE is non-derivable

The Yosida–Hewitt decomposition [12] identifies the obstruction Proposition 4.9 has exposed. Every finitely additive probability decomposes uniquely as $\ell = \ell_c + \ell_p$, where ℓ_c is σ-additive and ℓ_p is *purely finitely additive.* The finite-cofinite content has $\ell_p = \ell$. By Theorem 4.10, CE holds precisely when $\ell_p = 0$. The obstruction is therefore not a feature of any particular structural hypothesis: it is the persistence of a purely finitely additive component that no first-order structural condition can eliminate. Proposition 4.11 makes this precise.

Proposition 4.11 (CE non-derivability)**.** *Collective exhaustion is not implied by any structural condition on a query system.*

(i) (Particular case.) *Sequential upper-directedness, compatibility, and normalization do not imply CE.*

(ii) (Universal case.) *No condition Φ expressible in the first-order language of query systems implies CE.*

Proof. The particular case is Proposition 4.9: the finite-cofinite content satisfies all named structural hypotheses and fails CE.

For the universal case, we proceed in three steps.

Step 1. A first-order condition on query systems is preserved under ultraproducts by Łoś's theorem [8]: if Φ holds in every member of a family of query systems, it holds in their ultraproduct.

Step 2. Since the finite-cofinite construction works on any countably infinite set, we may realise it on $\mathbb{Q}$. Fix an enumeration $q : \mathbb{N} \to \mathbb{Q}$, let $\mathcal{U}$ be a nonprincipal ultrafilter on $\mathbb{N}$ extending the cofinite filter, and let δ_n be the point mass at $q(n) \in \mathbb{Q}$. Each δ_n is a normalized σ-additive content satisfying every structural condition including CE. The ultraproduct $\prod_{\mathcal{U}} \delta_n$ assigns to $E \subseteq \mathbb{Q}$ the value $\lim_{\mathcal{U}} \mathbf{1}_{q(n) \in E}$.

If E is finite then $q(n) \in E$ holds for only finitely many n, so the ultralimit is 0. If E is cofinite then $q(n) \in E$ holds for cofinitely many n, so the ultralimit is 1, since $\mathcal{U}$ extends the cofinite filter. The ultraproduct is therefore exactly the finite-cofinite content ℓ of Example 4.2 on $\mathbb{Q}$.

Step 3. By Step 1, any universally valid first-order condition Φ holds in the ultraproduct whenever it holds in each δ_n. But by Step 2 the ultraproduct is exactly ℓ, and ℓ fails CE. Therefore no first-order structural condition can imply CE. □

The ultrafilter-based content is the canonical witness: it passes every finite consistency test yet assigns unit mass to events whose infinite intersection is empty. CE is the admissibility condition under which observational coherence furnishes probability, not a structural consequence of the query system.

4.5 Finite coherence does not determine probability

The obstruction in Proposition 4.11 is not an artifact of the query-system language. In the more primitive setting of Boolean algebras with a normalized finitely additive charge, the same conclusion holds: no first-order theory in that language characterizes those algebras whose charge extends to a σ-additive measure. Equivalently, countable additivity is not first-order axiomatizable among finitely additive probability Boolean algebras [9]. The proof is the same ultraproduct mechanism: Dirac charges δ_n would satisfy any putative characterizing theory, yet their ultraproduct is the finite-cofinite charge ℓ, which fails σ-additivity — contradicting Łoś's theorem.

Finite coherence therefore underdetermines probability in a logically precise sense. Foundational disputes about probability are not terminological; they concern which extra structure licenses the passage from finite coherence to probability, and some such structure is unavoidable. CE is the condition adopted here; the companion note [9] gives the full general proof.

5 Stone realization from finitely additive data

The Stone construction shows that σ-additivity can instead be derived. Starting only from finitely additive charges, compactness of the Stone space produces a σ-additive measure on the compact completion; CE then reappears as the support condition that returns this measure to the realised instantiation space. Notably, no sequential common refinements are assumed: the σ-additivity that Carathéodory obtains by compressing countable covers within the index set is here produced by compactness of the Stone space, without any countable reach into ι.

5.1 Stone duality for the direct limit

Lemma 5.1 (Injectivity of connecting maps). *If the query system satisfies Surjective Evaluation, then $\varphi_{ij} : B_i \to B_j$ is an injective Boolean algebra homomorphism for every $i \leq j$.*

Proof. Surjective evaluation and the coherence condition give $\pi_{ij}(\mathrm{eval}_j(\omega)) = \mathrm{eval}_i(\omega)$ for all ω, so $\mathrm{im}(\pi_{ij}) \supseteq \mathrm{im}(\mathrm{eval}_i) = \mathsf{O}_i$: the refinement maps are surjective. A surjective map π_{ij} has injective preimage map π_{ij}^{-1} on subsets, so $\varphi_{ij}(\mathrm{Cyl}(i, A)) = \mathrm{Cyl}(j, \pi_{ij}^{-1}(A))$ is injective on generators, hence on B_i. $\square$

Proposition 5.2 (Stone duality for the direct limit). *The Stone space of the direct limit satisfies $\mathrm{St}\big(\bigcup_i B_i\big) \cong \varprojlim_i \mathrm{St}(B_i)$.*

Proof. By Lemma 5.1, the connecting maps φ_{ij} are injective Boolean algebra homomorphisms. Under Stone duality, injective Boolean algebra homomorphisms correspond to surjective continuous maps between Stone spaces. The Stone space functor St converts colimits in the category of Boolean algebras to limits in the category of compact Hausdorff spaces [6, II.4]. Applied to $\bigcup_i B_i$ with its injective system, this gives $\mathrm{St}\big(\bigcup_i B_i\big) \cong \varprojlim_i \mathrm{St}(B_i)$. $\square$

5.2 The inverse limit measure

Proposition 5.3 (Inverse limit measure). *Let $\{\mu_i\}$ be a compatible family of normalised charges on $\{B_i\}$. Then there exists a unique probability measure $\hat{\mu}$ on $\mathcal{B}(\varprojlim_i \mathrm{St}(B_i))$ whose pushforward along each projection $\varprojlim_i \mathrm{St}(B_i) \to \mathrm{St}(B_j)$ agrees with the Borel measure on $\mathrm{St}(B_j)$ corresponding to μ_j.*

Proof. Step 1: Single-algebra correspondence. For each i, Stone duality identifies B_i with the clopen algebra $\mathrm{Clop}(\mathrm{St}(B_i))$. A charge μ_i on B_i thus defines a finitely additive set function on the clopens of the compact Hausdorff space $\mathrm{St}(B_i)$. Since $\mathrm{St}(B_i)$ is compact and totally disconnected, the clopens form a base for the topology and a finitely additive charge on them extends uniquely to a regular Borel measure; see [2], §6, or [5], §§53–54. Call this measure $\hat{\mu}_i$ on $\mathrm{St}(B_i)$.

Step 2: Compatibility. The compatibility condition $\mu_j(\varphi_{ij}(E)) = \mu_i(E)$ translates, under Stone correspondence, to $\hat{\mu}_i$ being the pushforward of $\hat{\mu}_j$ along the bonding map $\mathrm{St}(B_j) \to \mathrm{St}(B_i)$. The family $\{\hat{\mu}_i\}$ is compatible on the inverse system $\{\mathrm{St}(B_i)\}$.

Step 3: Extension to the inverse limit. Each $\mathrm{St}(B_i)$ is compact Hausdorff. The inverse limit $\varprojlim_i \mathrm{St}(B_i)$ is compact Hausdorff. By [3, Theorem 1], a compatible family of regular Borel probability measures on a cofiltered inverse system of compact Hausdorff spaces with surjective bonding maps extends uniquely to a regular Borel probability measure on the inverse limit. This gives $\hat{\mu}$ on $\mathcal{B}(\varprojlim_i \mathrm{St}(B_i))$. $\square$

5.3 Structural identification

The Stone space $\mathrm{St}(\mathcal{C})$ contains Ω as a subspace via the Stone embedding, but only if distinct states are separated by observable events.

Definition 5.4 (Discriminability)**.** The query system *discriminates points* if for every $\omega \neq \omega'$ in Ω there exists $E \in \mathcal{C}$ with $\omega \in E$ and $\omega' \notin E$.

Discriminability ensures the Stone embedding is injective: without it, distinct states would be identified in the completion, and the pullback of the measure on $\mathrm{St}(\mathcal{C})$ would not recover a well-defined measure on Ω.

Let $\mathrm{pure} : \Omega \to \mathrm{St}(\mathcal{C})$ denote the Stone embedding, which sends $\omega \in \Omega$ to its principal ultrafilter $\mathrm{pure}(\omega) = \{E \in \mathcal{C} : \omega \in E\}$. For $E \in \mathcal{C}$, write $[E] = \{u \in \mathrm{St}(\mathcal{C}) : E \in u\}$ for the corresponding basic clopen in $\mathrm{St}(\mathcal{C})$.

Proposition 5.5 (Structural identification)**.** *Suppose the query system discriminates points. Then*

$$\mathrm{pure}^{-1}\big(\mathcal{B}_{\mathrm{Ba}}(\mathrm{St}(\mathcal{C}))\big) \; = \; \sigma(\mathcal{C}),$$

where $\mathcal{B}_{\mathrm{Ba}}$ denotes the Baire σ-algebra.

Proof. **($\supseteq$):** For any $E = \mathrm{Cyl}(i, A) \in \mathcal{C}$, $\mathrm{pure}^{-1}([E]) = \{\omega : E \in \mathrm{pure}(\omega)\} = \{\omega : \omega \in E\} = E$. Since $[E]$ is clopen, hence Baire, every cylinder set is in $\mathrm{pure}^{-1}(\mathcal{B}_{\mathrm{Ba}})$, and therefore $\sigma(\mathcal{C}) \subseteq \mathrm{pure}^{-1}(\mathcal{B}_{\mathrm{Ba}})$.

($\subseteq$): The Stone space $\mathrm{St}(\mathcal{C})$ is compact and totally disconnected, so its clopens form a topological basis. The clopens are exactly the sets $[E]$ for $E \in \mathcal{C}$ [6, I.3.2], so the Baire σ-algebra of $\mathrm{St}(\mathcal{C})$ is $\sigma(\{[E] : E \in \mathcal{C}\})$. For any Baire set V, $\mathrm{pure}^{-1}(V) \in \sigma(\{\mathrm{pure}^{-1}([E]) : E \in \mathcal{C}\}) = \sigma(\mathcal{C})$.

Injectivity. Discriminability (Definition 5.4) implies pure is injective: if $\omega \neq \omega'$, there exists $E \in \mathcal{C}$ with $\omega \in E$ and $\omega' \notin E$, so $E \in \mathrm{pure}(\omega)$ but $E \notin \mathrm{pure}(\omega')$. Injectivity ensures the pullback does not conflate distinct events. □

Remark 5.6. On a compact Hausdorff space the Baire σ-algebra is contained in the Borel σ-algebra, with equality when the space is second-countable. We work throughout with the Baire σ-algebra on $\mathrm{St}(\mathcal{C})$; this suffices because the measure $\hat{\mu}$ constructed above is a Baire measure (it is the unique extension of a charge on the clopen algebra), and Choksi's theorem applies to Baire measures on compact spaces.

5.4 CE as support condition

Proposition 5.7 (Support condition)**.** *Let μ be a charge on $\mathcal{C}$ satisfying CE. Then the corresponding Baire measure $\hat{\mu}$ on $\mathrm{St}(\mathcal{C})$ is supported on $\mathrm{pure}(\Omega)$, the set of principal ultrafilters.*

Proof. By the Yosida–Hewitt theorem [12, 10], every bounded charge on a Boolean algebra decomposes uniquely as $\mu = \mu_c + \mu_p$, where μ_c is countably additive and μ_p is purely finitely additive. Under Stone duality, μ_c corresponds to mass on the principal ultrafilters $\mathrm{pure}(\Omega) \subset \mathrm{St}(\mathcal{C})$, while μ_p corresponds to mass on the non-principal ultrafilters [10], Ch. 10.

CE is equivalent to $\mu_p = 0$: CE requires $\mu(E_n) \to 0$ whenever $E_n \searrow \emptyset$, which is exactly the condition that the purely finitely additive part vanishes (Theorem 4.10). Therefore $\hat{\mu}_p = 0$ and $\hat{\mu}$ is supported on $\mathrm{pure}(\Omega)$. □

5.5 The Stone observational extension

Theorem 5.8 (Stone observational extension)**.** *Let the query system satisfy Surjective Evaluation, Discriminability, and common coarsenings, and let $\{\mu_i\}$ be a compatible family of normalised charges on $\{B_i\}$ satisfying CE. Then there exists a unique probability measure P on $(\Omega, \sigma(\mathcal{C}))$ such that*

$$P(\mathrm{Cyl}(i, A)) \;=\; \mu_i(\mathrm{Cyl}(i, A)) \qquad \textit{for all } i \in \iota \textit{ and } A \in \mathcal{O}_i.$$

Proof. The proof assembles the preceding propositions:

$$\{\mu_i\} \xrightarrow{\S 5.2,\,\text{Step 1}} \{\hat{\mu}_i\} \xrightarrow{\S 5.2,\,\text{Step 3}} \hat{\mu} \text{ on } \varprojlim_i \mathrm{St}(B_i) \xrightarrow{\text{Prop. 5.5}} P \text{ on } \sigma(\mathcal{C}).$$

Existence. By Proposition 5.2, $\mathrm{St}(\mathcal{C}) \cong \varprojlim_i \mathrm{St}(B_i)$. By Proposition 5.3, the compatible family $\{\mu_i\}$ extends to a Baire probability measure $\hat{\mu}$ on $\varprojlim_i \mathrm{St}(B_i) \cong \mathrm{St}(\mathcal{C})$.

By Proposition 5.7, $\hat{\mu}$ is supported on $\mathrm{pure}(\Omega)$. Since pure is injective (discriminability), the pushforward $P = (\mathrm{pure})_*\hat{\mu}$ — the image measure under pure^{-1} on the support — is a well-defined probability measure on Ω.

By Proposition 5.5, pure^{-1} maps Baire sets in $\mathrm{St}(\mathcal{C})$ to sets in $\sigma(\mathcal{C})$, so P is defined on $\sigma(\mathcal{C})$. For any cylinder set $E = \mathrm{Cyl}(i, A)$,

$$P(E) \;=\; \hat{\mu}([E]) \;=\; \mu_i(\mathrm{Cyl}(i, A)).$$

Uniqueness. Two probability measures on $(\Omega, \sigma(\mathcal{C}))$ agreeing on all cylinder sets are equal by Theorem 3.1, which requires common coarsenings. The Stone theorem therefore assumes common coarsenings in addition to surjective evaluation, discriminability, and CE. □

It may appear that compactness of the Stone space eliminates the need for CE. Compactness ensures $\hat{\mu}$ on $\mathrm{St}(\mathcal{C})$ is σ-additive, but $\hat{\mu}$ lives on the Stone space, not on Ω. Descent to Ω requires that no mass escapes to non-principal ultrafilters, and that is precisely CE (Proposition 5.7). In the Stone picture, CE is not an additional hypothesis but a *support condition*: the measure concentrates on the image of Ω inside its Stone compactification.

6 Synthesis

The Yosida–Hewitt decomposition identifies the algebraic and geometric faces of CE: $\mu_p = 0$ is both the vanishing of the purely finitely additive part and the condition that no mass escapes to non-principal ultrafilters.

When the index set admits sequential common refinements and the query system discriminates points, both constructions are available simultaneously. The following proposition confirms they produce the same observable measure.

Proposition 6.1 (Uniqueness of the observable measure)**.** *Suppose the query system satisfies Surjective Evaluation, Discriminability, and sequential common refinements and common coarsenings, and let $\{\nu_i\}$ be a compatible family of probability measures. Let P^{C} be the measure produced by Theorem 3.5 and P^{S} the measure produced by Theorem 5.8. Then $P^{\mathrm{C}} = P^{\mathrm{S}}$.*

Proof. Both measures agree on every cylinder set: $P^{\mathrm{C}}(\mathrm{Cyl}(i, A)) = \nu_i(A) = P^{\mathrm{S}}(\mathrm{Cyl}(i, A))$. By observational determination (Theorem 3.1), they are equal on $\sigma(\mathcal{C})$. □

Corollary 6.2. *The observable measure may be obtained by two distinct constructions: by countable reach in ι via sequential common refinements (Theorem 3.5), or by the Stone completion of $\mathcal{C}$ (Theorem 5.8). Whenever both constructions apply, they yield the same measure.*

The Stone space makes the geometry of that demand visible. Every point of $\mathrm{St}(\mathcal{C})$ is a maximal consistent assignment of yes/no answers to every observable distinction — a coherent distinction pattern, whether or not any $\omega \in \Omega$ realises it. The observer's world sits inside $\mathrm{St}(\mathcal{C})$ as the principal ultrafilters; beyond it lie ideal limit points, internally coherent but unrealised.

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